



SQL / PostgreSQL

Indexes · MVCC · Window Fns · Partitioning · Replication

Overview

PostgreSQL-specific features: advanced indexing, MVCC, window functions, stored procedures/triggers, table partitioning, streaming replication, and cross-dialect compatibility.

Index Types Used

Index Type	SQL	Use Case
B-Tree	idx_orders_customer ON orders(customer_id)	Equality & range on customer_id
Partial	idx_orders_pending WHERE status='PENDING'	Filter-heavy queries on pending orders
Covering	idx_orders_cov INCLUDE (status,total,created_at)	Index-only scan for order list
Composite	idx_order_items_comp(order_id, product_id)	Order-item join queries
GIN	idx_products_tags USING GIN(tags)	Array @> containment queries
Expression	idx_products_lower_name(LOWER(name))	Case-insensitive search

Transaction Isolation Levels

Level	Used In	Protects Against	Cost
READ COMMITTED	Default	Dirty reads	Low
REPEATABLE READ	placeOrder()	Dirty + non-repeatable reads	Medium
SERIALIZABLE	generateDailyReport()	All anomalies incl. phantom reads	High; use @Retryable

MVCC note: PostgreSQL uses xmin/xmax per row tuple; no lock needed for reads. Autovacuum reclaims dead tuples – monitor n_dead_tup via pg_stat_user_tables.

Advanced SQL Features

Feature	Query	Class
Window ROW_NUMBER	OVER (PARTITION BY customer_id ORDER BY total DESC)	CustomerOrderValueQuery
LATERAL JOIN	3 most recent orders per customer in one query	RecentOrdersPerCustomerQuery
LAG / LEAD	Month-over-month revenue change	RevenueReport
Running SUM	SUM(total) OVER (ORDER BY created_at)	Monthly revenue trend
Anti-join NOT EXISTS	Products never ordered	ProductRepository native query
CTE	Monthly revenue rollup	OrderStatisticsRepository
UPSERT	INSERT ... ON CONFLICT (sku) DO UPDATE	ProductSyncService

Stored Procedures & Triggers (V4–V6)

sp_complete_order(order_id) – atomic: update status + decre

fn_customer_lifetime_value(customer_id) – returns NUMERIC

trg_orders_audit – AFTER INSERT/UPDATE/DELETE → order_audit

trg_prevent_status_regression – BEFORE UPDATE raises EXCEPT

Partitioning & Replication

Feature	Detail	Class
PARTITION BY RANGE(create_date)	Monthly partitions on orders table	V7__partitioning.sql
orders_default partition	Catches rows outside defined ranges	V7__partitioning.sql
PartitionManagerJob	@Scheduled(cron) creates next month's partition	PartitionManagerJob
RoutingDataSource	Writes → primary, reads → replica	DataSourceConfig
ReadWriteRoutingAspect	@Around sets "read"/"write" context	ReadWriteRoutingAspect
ReplicationLagHealthIndicator	Queries pg_stat_replication; alerts if lag > 50ms	HealthIndicator